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The CCDI-COG Molecular Characterization Initiative (MCI) in Pediatric Rare Cancers

Rare pediatric cancers encompass a broad range of diagnoses, including both adult-onset carcinomas that rarely arise in children and very rare diagnoses unique to childhood.

Although each individual diagnosis is exceedingly uncommon, rare tumors collectively

account for ~10% of all children with cancer. However, for many of these cancers, little is known about their genetic drivers or optimal treatment. The Molecular Characterization Initiative (MCI), launched in 2022 by the NCI Childhood Cancer Data Initiative (CCDI), provides molecular profiling at no cost to the patients.

Individuals 25 years old or younger newly diagnosed with a rare tumor, as defined by the Children's Oncology Group (COG), are eligible for paired germline and tumor enhanced exome sequencing of cancer-associated genes, copy number analysis through low-pass whole genome sequencing, and targeted RNA fusion analysis. Results are returned to the primary institution within 2-3 weeks.

From 9/22/2022 to 03/31/2026, 872 individuals were enrolled from 160 institutions across 6 countries. The most common diagnosis subgroups were thyroid carcinoma (n=204), neuroendocrine tumors (n=92), sex cord-stromal tumors (n=75), and other rare tumors (n=182). Of the 683 patients with tumor/germline sequencing results, clinically actionable (Tier I/II) germline single nucleotide variants (SNV) and/or copy number variants (CNV) were identified in 153 patients (22.4%). The most prevalent germline alterations affected DICER1 (n=37, 5.4%), TP53 (n=16, 2.3%), RB1 (n=13, 1.9%), and VHL (n=11, 1.6%). Germline variants were generally consistent with established genotype-tumor associations, though several patients had large copy number variants that would have been challenging to detect with conventional targeted sequencing panels. In addition, 52.3% (n=357) of tumors demonstrated a Tier I/II SNV somatic variant and 54.3% (n=371) had a somatic CNV. Novel associations included KMT2D truncation and SMARCA4 missense variants, each seen in 4 children with nasopharyngeal carcinoma. Of the 583 patients with fusion results, oncogenic fusions were identified in 130 patients (22.8%), most commonly associated with thyroid cancer (affecting RET, NTRK3, and other kinases), desmoplastic small round cell tumor (EWSR1::WT1), or mucoepidermoid carcinoma (CRTC1::MAML2). Novel fusions included ADGRG12::NOTCH2 (adenoid cystic carcinoma), ERC1::BRAF (pancreatoblastoma), and RB1::DIAPH3 (retinoblastoma). Among the 229 patients with follow-up data, 21 (9.2%) were reported to receive a therapy matched to a molecular alteration identified by MCI.

The MCI has improved access to tumor genomic profiling for a wide range of rare tumors across international COG centers. Germline cancer predisposition was identified in 22.4% of individuals with rare childhood cancers. These results highlight the importance of molecular profiling for the purposes of genetic counseling, tailoring therapy, and improving our understanding of the genetic drivers of rare pediatric tumors.